

# “Calculus 3”

## Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

### Day 20

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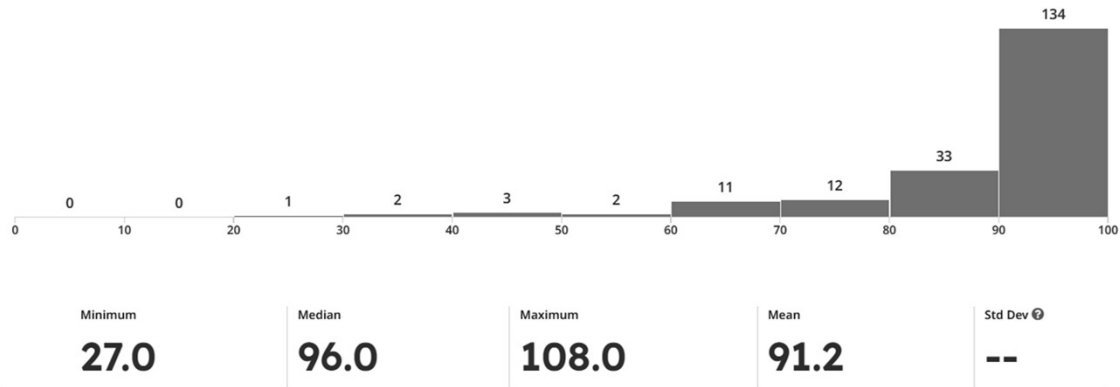
### Any Reminders? Any Questions?

- Quiz on Friday (discussion) covers 16.2 (line integrals)
- Office hours Mondays 1-2pm online
- Office hours Thursday 3:30-4:30 at MONT 233
- Calc Night Thursday 6:30-8:30 at MONT 104

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# Exam 2: Stats!



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*ALVARO:* Start the recording!

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# “Calculus 3”

## Multi-Variable Calculus

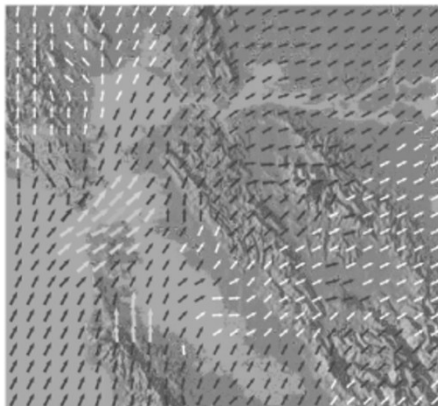
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## Vector Fields

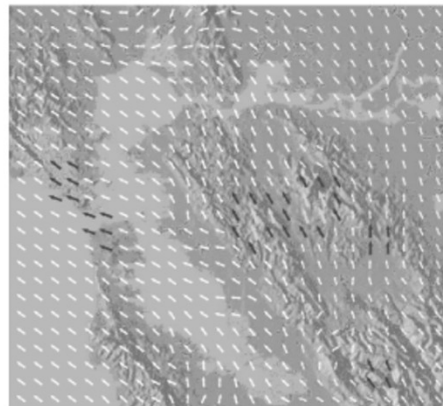
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### Vector Fields

Velocity vector fields showing San Francisco Bay wind patterns on a particular spring day



(a) 6:00 PM



(b) 6:00 AM

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# Vector Fields

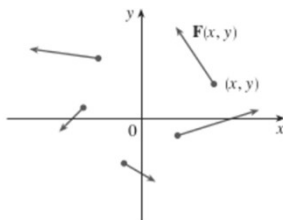


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## Vector Fields

Let  $D$  be a set in  $\mathbb{R}^2$  (a plane region). A **vector field on  $\mathbb{R}^2$**  is a function  $\mathbf{F}$  that assigns to each point  $(x, y)$  in  $D$  a two-dimensional vector  $\mathbf{F}(x, y)$ .

Let  $E$  be a subset of  $\mathbb{R}^3$ . A **vector field on  $\mathbb{R}^3$**  is a function  $\mathbf{F}$  that assigns to each point  $(x, y, z)$  in  $E$  a three-dimensional vector  $\mathbf{F}(x, y, z)$ .

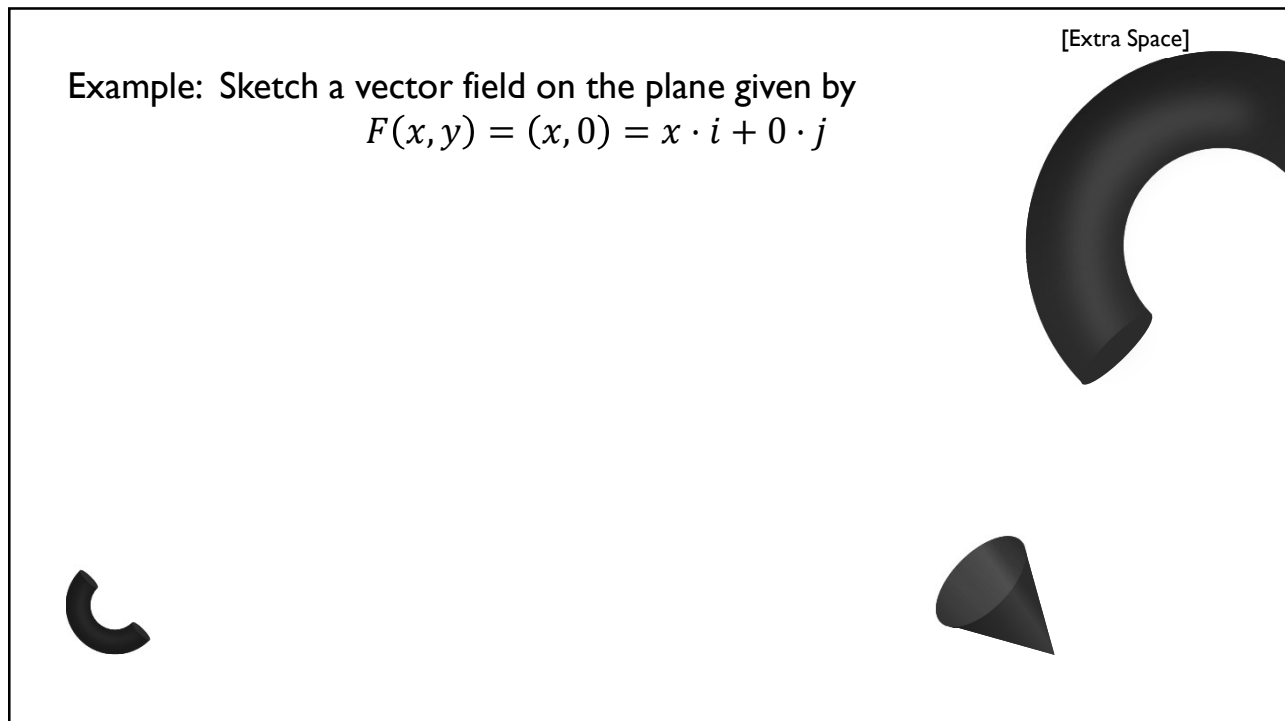


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Example: Sketch a vector field on the plane given by

$$F(x, y) = (x, 0) = x \cdot i + 0 \cdot j$$

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Example: Sketch a vector field on the plane given by

$$F(x, y) = (-y, x) = -y \cdot i + x \cdot j$$



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Example: Sketch a vector field on the plane given by

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[Extra Space]



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Example: Sketch the gravitational vector field of the gravity force exerted on a planet of mass  $m$  by a star of mass  $M$  (located at the origin).  
 (Hint: recall that Newton's Laws of Gravitation say that the magnitude of the force is given by  $mMG/r^2$  .)

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[Extra Space]

Example: Sketch the gravitational vector field of the gravity force exerted on a planet of mass  $m$  by a star of mass  $M$  (located at the origin).

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## The Gradient Vector Field

If  $f$  is a function of two variables  $x$  and  $y$ , then the **gradient** of  $f$  is the vector function  $\nabla f$  defined by

$$\nabla f(x, y) = \langle f_x(x, y), f_y(x, y) \rangle = \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j}$$

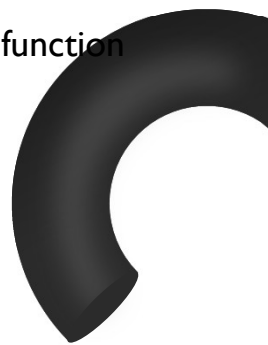
$$\nabla f(x, y) = f_x(x, y) \mathbf{i} + f_y(x, y) \mathbf{j}$$



$$\nabla f(x, y, z) = f_x(x, y, z) \mathbf{i} + f_y(x, y, z) \mathbf{j} + f_z(x, y, z) \mathbf{k}$$

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Example: Sketch the gradient vector field on the plane for the function

$$f(x, y) = x^2 + y^2$$


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[Extra Space]

Example: Sketch the gradient vector field on the plane for the function

$$f(x, y) = x^2 + y^2$$

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Example: Let  $F(\vec{x}) = -\frac{mMG}{|\vec{x}|^3} \vec{x}$  be a gravitational vector field (in 3D), and let

$$f(\vec{x}) = \frac{mMG}{|\vec{x}|} .$$

Show that  $F = \nabla f$  .

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[Extra Space]

Example: Let  $F(\vec{x}) = -\frac{mMG}{|\vec{x}|^3} \vec{x}$  be a gravitational vector field (in 3D), and let  $f(\vec{x}) = \frac{mMG}{|\vec{x}|}$ . Show that  $F = \nabla f$ .

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## The Gradient Vector Field

If  $f$  is a function of two variables  $x$  and  $y$ , then the **gradient** of  $f$  is the vector function  $\nabla f$  defined by

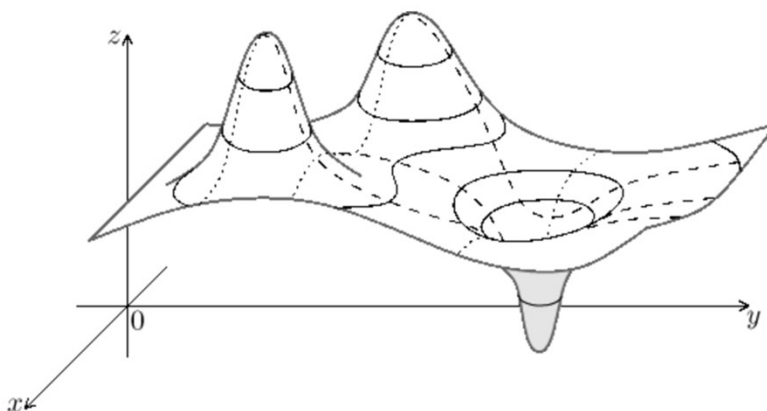
$$\nabla f(x, y) = \langle f_x(x, y), f_y(x, y) \rangle = \frac{\partial f}{\partial x} \mathbf{i} + \frac{\partial f}{\partial y} \mathbf{j}$$

### 15 Theorem

Suppose  $f$  is a differentiable function of two or three variables. The maximum value of the directional derivative  $D_{\mathbf{u}}f(\mathbf{x})$  is  $|\nabla f(\mathbf{x})|$  and it occurs when  $\mathbf{u}$  has the same direction as the gradient vector  $\nabla f(\mathbf{x})$ .

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## The Gradient Vector Field and “Gradient Descent” on “loss functions” in the training of AI/LLM models



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## Line Integrals of Vector Fields: An Example

Recall that the “work” done by a constant force  $F$  (in vector form) in moving an object from point  $P$  to point  $Q$  in space is given by

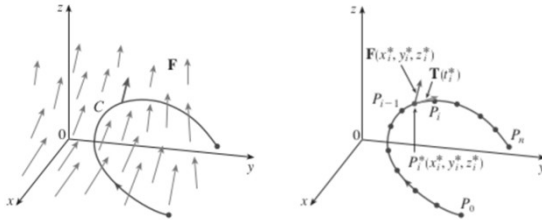
$$W = F \cdot D$$

where  $D = \overrightarrow{PQ}$ .

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## Line Integrals of Vector Fields: An Example

What is the “work” done by a force  $\mathbf{F}$  (in vector form) in moving an object from point  $P$  to point  $Q$  in space as we travel over a curve  $\mathbf{r}(t)$ ?



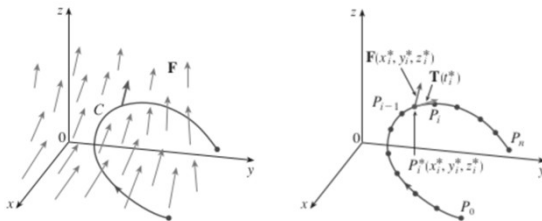
$$\sum_{i=1}^n [\mathbf{F}(x_i^*, y_i^*, z_i^*) \cdot \mathbf{T}(x_i^*, y_i^*, z_i^*)] \Delta s_i$$

$$W = \int_C \mathbf{F}(x, y, z) \cdot \mathbf{T}(x, y, z) ds = \int_C \mathbf{F} \cdot \mathbf{T} ds$$

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## Line Integrals of Vector Fields: An Example

What is the “work” done by a force  $\mathbf{F}$  (in vector form) in moving an object from point  $P$  to point  $Q$  in space as we travel over a curve  $\mathbf{r}(t)$ ?



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$$W = \int_C \mathbf{F}(x, y, z) \cdot \mathbf{T}(x, y, z) ds = \int_C \mathbf{F} \cdot \mathbf{T} ds$$

$$W = \int_a^b \left[ \mathbf{F}(\mathbf{r}(t)) \cdot \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|} \right] |\mathbf{r}'(t)| dt = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt$$

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Example: Find the work done by the force

$$F(x, y) = x^2 \cdot i - xy \cdot j$$

in moving a particle along the quarter circle from (1,0) to (0,1).



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[Extra Space]

Example: Find the work done by the force

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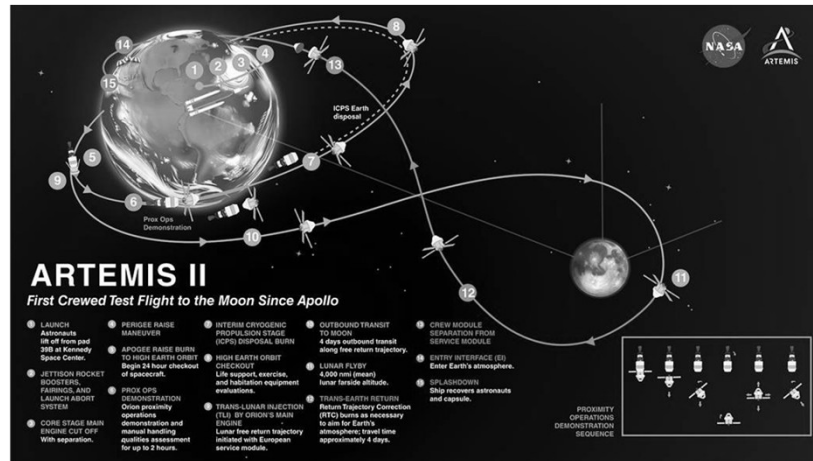
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## Line Integrals of Vector Fields: An Example

“Work” represents the amount of energy necessary to move an object in a particular path...



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Example: Set up an integral that describes the work done by gravitational forces on Artemis II during its path  $r(t)$ .

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## Line Integrals of Vector Fields in General

Let  $\mathbf{F}$  be a continuous vector field defined on a smooth curve  $C$  given by a vector function  $\mathbf{r}(t)$ ,  $a \leq t \leq b$ . Then the **line integral of  $\mathbf{F}$  along  $C$**  is

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt = \int_C \mathbf{F} \cdot \mathbf{T} ds$$

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Example: Evaluate  $\int_C \mathbf{F} \cdot d\mathbf{r}$  where

$$\mathbf{F}(x, y, z) = xy \cdot \mathbf{i} + yz \cdot \mathbf{j} + xz \cdot \mathbf{k}$$

and  $C$  is the “twisted cubic” given by  $\mathbf{r}(t) = (t, t^2, t^3)$ ,  $0 \leq t \leq 1$ .

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[Extra Space]

Example: Evaluate  $\int_C \mathbf{F} \cdot d\mathbf{r}$  where

$$\mathbf{F}(x, y, z) = xy \cdot \mathbf{i} + yz \cdot \mathbf{j} + xz \cdot \mathbf{k}$$

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## Line Integrals of Vector Fields in General

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$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt = \int_C \mathbf{F} \cdot \mathbf{T} ds$$

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C P dx + Q dy + R dz \quad \text{where } \mathbf{F} = P \mathbf{i} + Q \mathbf{j} + R \mathbf{k}$$

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## Line Integrals of Vector Fields in General

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C P dx + Q dy + R dz \quad \text{where } \mathbf{F} = P \mathbf{i} + Q \mathbf{j} + R \mathbf{k}$$

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Questions?

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ALVARO: Start the recording!



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“Calculus 3”



Multi-Variable Calculus

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The Fundamental Theorem  
for Line Integrals



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# The Fundamental Theorem of Calculus

## The Fundamental Theorem of Calculus, Part 2

If  $f$  is continuous on  $[a, b]$ , then

$$\int_a^b f(x) \, dx = F(b) - F(a)$$

where  $F$  is any antiderivative of  $f$ , that is, a function  $F$  such that  $F' = f$ .

$$\int_a^b F'(x) \, dx = F(b) - F(a)$$

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# The Fundamental Theorem for Line Integrals

## 2 Theorem

Let  $C$  be a smooth curve given by the vector function  $\mathbf{r}(t)$ ,  $a \leq t \leq b$ . Let  $f$  be a differentiable function of two or three variables whose gradient vector  $\nabla f$  is continuous on  $C$ . Then

$$\int_C \nabla f \cdot d\mathbf{r} = f(\mathbf{r}(b)) - f(\mathbf{r}(a))$$

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## The Fundamental Theorem for Line Integrals: Proof

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## The Fundamental Theorem for Line Integrals: Proof

$$\begin{aligned}
 \int_C \nabla f \cdot d\mathbf{r} &= \int_a^b \nabla f(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt \\
 &= \int_a^b \left( \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} + \frac{\partial f}{\partial z} \frac{dz}{dt} \right) dt \\
 &= \int_a^b \frac{d}{dt} f(\mathbf{r}(t)) dt \quad (\text{by the Chain Rule}) \\
 &= f(\mathbf{r}(b)) - f(\mathbf{r}(a))
 \end{aligned}$$

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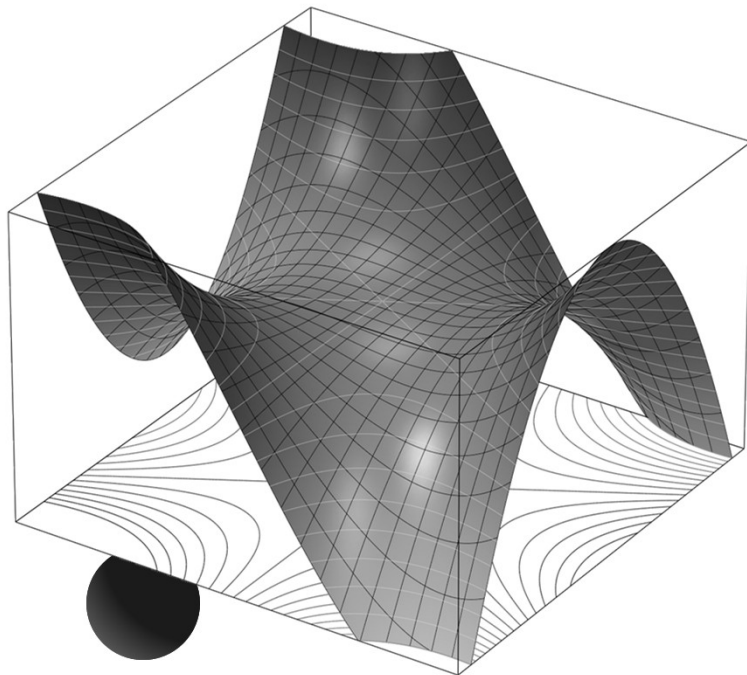
Questions?



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Thank you

Until next time.



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